$PM_{2.5}$ Forecasting During Smog Seasons Using Time Series Analysis

DS207: Time Series Forecasting

15th of May, 2026

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***Abstract*—Air pollution is one of the most dangerous public health challenges globally, and one of its pollutants, $PM_{2.5}$, is the most health-threatening one. This project focuses on forecasting $PM_{2.5}$ levels during the smog seasons using time series forecasting techniques. [1], [2] The Global Air Pollution Dataset (2025-2026) was obtained from Kaggle, which contains hourly pollutant observations from major smog cities such as Lahore, Beijing, London, and others. After preprocessing and feature engineering procedures, the models SARIMAX (recursive and dynamic), VARMAX, and XGBoost were applied alongside a Naive (last value) baseline to conduct interpretable comparisons between them and select the best one for the forecasts. Evaluation metrics chosen to quantify the performance of the models included MAE, RMSE, MAPE, and sMAPE. The main modeling workflow focuses on Lahore with a 48-hour forecasting window. The final analysis concluded recursive SARIMAX as the best model that achieved the lowest MAPE value of 6.108%, and further, it was selected for the final 7-day forecasts (the third week of February 2026) for all the cities mentioned in the dataset. Furthermore, the forecasts were validated using the actual observations retrieved from the WAQI platform, concluding the overall robustness and reliability of SARIMAX forecasts and depicting the final results using the MAPE score for all the cities separately.**

***Index Terms*—Air Pollution Forecasting, Time Series Forecasting, SARIMAX, VARMAX, XGBoost, Smog Analysis**

I. Introduction

Air pollution is a major public health issue that affects millions of people worldwide. Of all the pollutants, including SO_2, NO_2, and CO, $PM_{2.5}$ is particularly harmful. Its small particles can reach deep into the lungs and bloodstream, leading to serious respiratory and heart problems. [6] During smog seasons, $PM_{2.5}$ levels often increase because of traffic, industrial emissions, and certain weather conditions.

Hence, it is extremely important to have accurate $PM_{2.5}$ forecasts for public protection and environmental monitoring. These forecasts enable governmental organizations to take preventive measures, such as reducing outdoor exposure or implementing some traffic restrictions.

Therefore, the objective of this project is to analyze the pollution behavior carefully and generate reliable short-term forecasts for highly polluted cities around the world. In this matter, time series and machine learning approaches can suggest satisfactory predictive models [1], [2], [4] to capture daily recurring patterns and pollution spikes. For such a motive, several forecasting models were implemented and evaluated using different error metrics.

II. Dataset Description

TABLE I
Summary of the Global Air Pollution Dataset

Feature	Type	Description
Date	DateTime	Hourly timestamps capturing temporal behavior and daily seasonality.
City	Categorical	Cities included in the study such as Lahore, Delhi, Beijing, London, and New York.
$PM_{2.5}$	Float	Target variable representing fine particulate matter concentration.
PM_{10}	Float	Coarse particulate matter concentration.
NO_2	Float	Nitrogen dioxide concentration associated with traffic and industrial emissions.
SO_2	Float	Sulfur dioxide concentration related to fossil fuel combustion.
CO	Float	Carbon monoxide concentration correlated with combustion activities.
Ozone	Float	Secondary pollutant influencing smog formation.
AOD	Float	Aerosol Optical Depth measuring atmospheric haziness.
AQI_Class	Category	Air quality classification ranging from Good to Hazardous.

The *Global Air Pollution Dataset (2025-2026)* was obtained from the open-source `Kaggle` website [5] and contains $17,472$ hourly city-wise pollutant observations. The dataset is multivariate, containing several atmospheric pollutants, including NO_2, SO_2, CO, $Ozone$, and PM_{10}. Additionally, it contained features such as AQI_Class a class from 'good' to 'hazardous' used to describe the threshold when any of the pollutants becomes dangerous for health and AOD a measure of atmospheric haziness; these features were mainly used in the EDA part but later excluded as they gave no value to the forecasting process.

During the first phase of the time series analysis, an exploratory analysis, strong daily seasonality, recurring smog cycles, and correlations among pollutants were observed. Here, PM_{10} was removed from the further experiments because of its extremely high positive correlation with $PM_{2.5}$, which introduced multicollinearity, and Ozone was also removed due to a negative correlation with the target $PM_{2.5}$. The cleaned $PM_{2.5}$ dataset was created during preprocessing to handle missing values and filter outliers for forecasting cities included in the dataset, such as `Beijing`, `Delhi`, `Dhaka`, `Jakarta`, `Lahore`, `London`, `New York`, and `Sao Paulo`.

III. Exploratory Time Series Analysis

Initial exploratory analysis started with analyzing the pollutants with respect to hourly observations in Lahore. Figure 1 presents short-term fluctuations across $PM_{2.5}$, PM_{10}, NO_2, SO_2, CO, and Ozone. These recurring fluctuations confirm that during smog seasons, pollution behavior is highly dynamic.

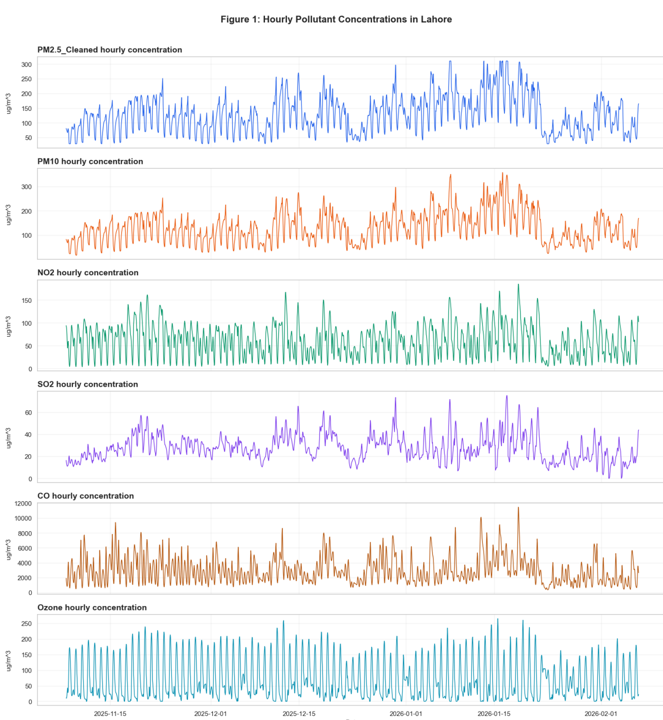

Fig. 1. Hourly pollutant concentrations in Lahore.

To determine whether pollutant behavior is a long-term indicator of a smog season, daily average concentrations were also analyzed, as shown in Figure 2. Compared to hourly observations, daily observations appear smoother because the time horizon is longer, and trends and recurring seasonal patterns are more clearly observable. $PM_{2.5}$ and PM_{10} show nearly identical patterns, visually indicating a very strong positive correlation.

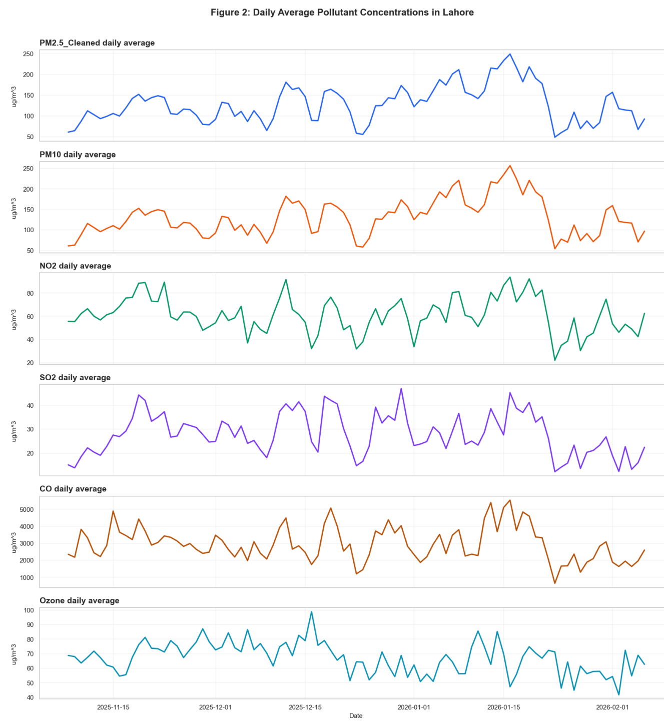

Fig. 2. Daily average pollutant concentrations in Lahore.

Figure 3 exemplifies normalized pollutant interactions during a smog week in Lahore. $PM_{2.5}$, PM_{10}, NO_2, SO_2, and CO generally progress together over time, suggesting possible interdependencies among these pollutants. In contrast, Ozone frequently follows opposite movement patterns in relation to $PM_{2.5}$.

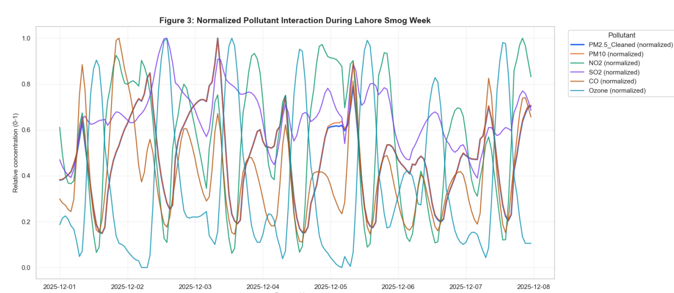

Fig. 3. Normalized pollutant interaction during a Lahore smog week.

The correlation heatmap in Figure 4 confirms these relationships. PM_{10} exhibits a perfect positive correlation of 1.00 with $PM_{2.5}$. NO_2, SO_2, and CO also show a strong correlation with the target variable, but a much more moderate one. Ozone demonstrates a moderate negative correlation with $PM_{2.5}$; thus, it was removed from the further modeling stages as it causes an inverse relationship with the target. Likewise, PM_{10} was removed due to causing multicollinearity with the target.

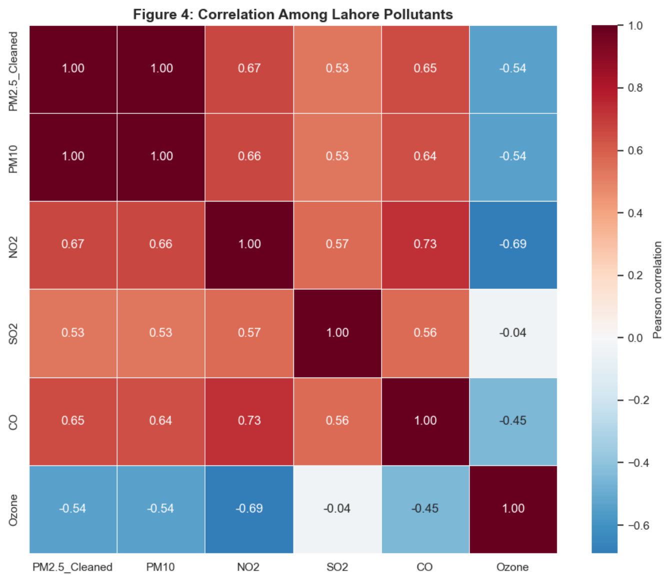

Fig. 4. Correlation matrix among Lahore pollutants.

Hourly $PM_{2.5}$ distribution boxplots are shown in Figure 5. Higher $PM_{2.5}$ levels are commonly observed during morning and evening hours, and the lowest concentration levels appear during the midday hours. This behavior indicates a strong intra-day seasonality likely associated with certain traffic activity or atmospheric conditions.

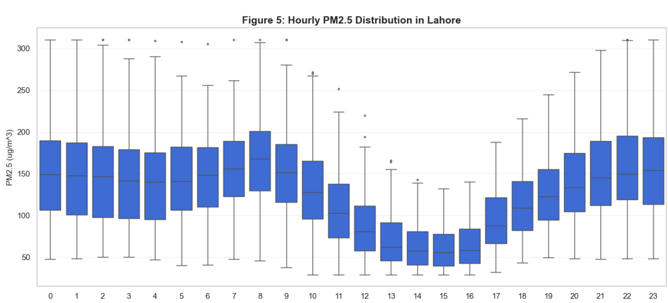

Fig. 5. Hourly $PM_{2.5}$ distribution in Lahore.

Seasonal decomposition of the cleaned $PM_{2.5}$ series is presented in Figure 6. The decomposition reveals a strong repetitive 24-hour seasonality, which can suggest that $PM_{2.5}$ contains significant temporal dependencies that the analysis will be focused on uncovering further.

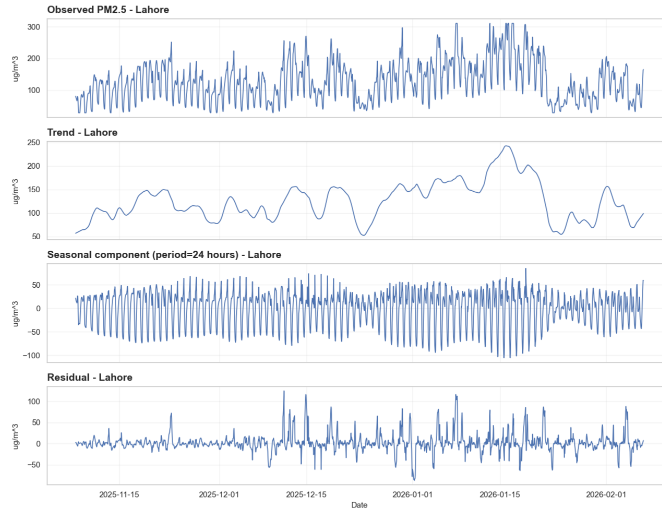

Fig. 6. Seasonal decomposition of cleaned $PM_{2.5}$ series.

Finally, the ACF and PACF analyses shown in Figure 7 indicate significant autocorrelation across multiple lags, particularly around the 24-hour seasonal interval, confirming the presence of strong seasonality in the $PM_{2.5}$ series.

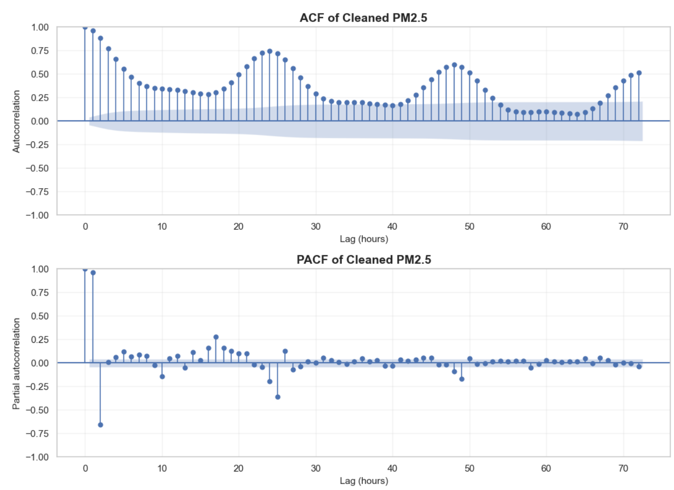

Fig. 7. ACF and PACF analysis of cleaned $PM_{2.5}$ series.

Overall, the exploratory analysis of this time series demonstrated that the dataset exhibits strong seasonality and temporal correlation, making it suitable for future time-series forecasting methods.

IV. Methodology

A. Data Preprocessing

Several preprocessing techniques were applied before training, including converting date columns to `datetime` format, resampling to an hourly frequency, handling missing values by forward-filling, filtering outliers using percentile-based clipping, and creating a cleaned version of the $PM_{2.5}$ series. These steps ensured a consistent temporal ordering and minimized the effect of extreme anomalies.

B. Feature Engineering

During the feature engineering process, several features were created involving: hour of day, day of week, lag features for $PM_{2.5}$, NO_2, SO_2, and CO, as well as rolling mean and standard deviation.

C. Forecasting Models

1) SARIMAX: SARIMAX is used to model $PM_{2.5}$ as a seasonal autoregressive integrated moving average process with exogenous pollutant features. The choice of SARIMAX was obvious, as shown in previous analyses, given that the data have strong seasonality and exogenous pollutants such as NO_2, SO_2, and CO that are collinear with the target $PM_{2.5}$. The selected model specification is `SARIMAX (1, 0, 2)` with seasonal order `(1, 1, 1, 24)`, selected by the lowest AIC of 2843.19.

Two SARIMAX forecasting modes were evaluated: the $recursive$ and $dynamic$ modes. The *recursive* mode predicts one step ahead and updates the model state with the latest actual observation during the holdout. The *dynamic* mode forecasts further ahead without the same stepwise actual update behavior.

TABLE II
SARIMAX HOLDOUT FORECASTS

Date	Actual	Recursive	Dynamic
2026-01-19 19:00	172.8	173.26	173.26
2026-01-19 21:00	204.0	202.34	201.38
2026-01-19 23:00	230.0	222.64	218.32
2026-01-20 02:00	237.5	237.14	210.17

2) VARMAX: VARMAX models multiple pollutant series jointly. The variables include $PM_{2.5}\_cleaned$, NO_2, SO_2, and CO. The stationarity check was re-performed to confirm the series is stationary, and the best VARMAX order, `VARMAX (2,0)`, was again selected based on the lowest AIC. All the other pollutants *Granger cause* the target $PM_{2.5}$ to be affected by. VARMAX is useful because it can capture interactions among pollutants, rather than treating non-target pollutants solely as external regressors, as analyzed previously when pollutants interact strongly with each other.

TABLE III
VARMAX HOLDOUT FORECASTS

Date	Actual	VARMAX
2026-01-19 19:00	172.8	168.20
2026-01-19 21:00	204.0	199.91
2026-01-19 23:00	230.0	214.40
2026-01-20 02:00	237.5	236.41

3) XGBoost: XGBoost is implemented as a supervised regression model. [4] It uses engineered time-series features such as $PM_{2.5}\_lag\_1$, $PM_{2.5}\_lag\_24$, rolling $PM_{2.5}$ statistics, pollutant features, and time features. The forecasting mode is recursive, making the method flexible but sensitive to accumulated forecast error.

TABLE IV
XGBOOST HOLDOUT FORECASTS

Date	Actual	XGBoost
2026-01-19 19:00	172.8	175.74
2026-01-19 21:00	204.0	202.06
2026-01-19 23:00	230.0	217.66
2026-01-20 02:00	237.5	209.61

V. Experimental Setup

Since time series forecasting relies on historical data, the dataset was split into training and test sets without shuffling. The main analyses were conducted on Lahore, training the forecasting models using historical pollutant observations, and were evaluated on a holdout forecast with a 48-hour window. The decision to focus on Lahore was random, and the main idea was not to overload the training runs by analyzing all cities at once. Instead, choose one, get results, identify which model is the best, and apply it for forecasting the remaining cities.

Several forecasting models were implemented, including `Naive Baseline`, `SARIMAX` in recursive and dynamic modes, `VARMAX`, and `XGBoost`. The recursive forecasting mode was used for multi-step prediction, where the previously forecasted values are iteratively used to generate future updates. Multivariate models incorporated lagged pollutant variables and engineered temporal features to improve forecasting performance.

Model evaluation was measured using several metrics [1] such as *Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), Symmetric Mean Absolute Percentage Error (sMAPE)*, and the *Coefficient of Determination (R^2)*. Among these, `MAPE` was selected to be the primary criterion for indicating the performance measure, as it provides interpretable error comparisons in percentages across models.

Figure 8 illustrates the 48-hour $PM_{2.5}$ forecasting experiment comparing SARIMAX recursive and SARIMAX dynamic forecasts against the actual $PM_{2.5}$ observations.

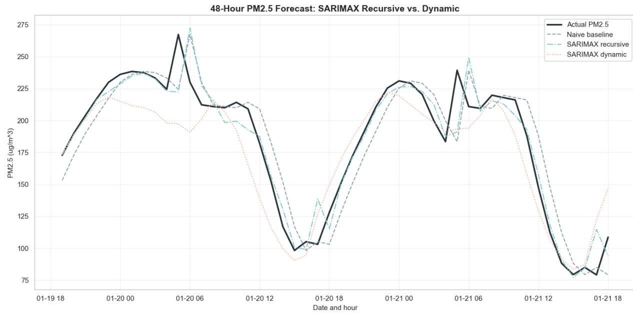

Fig. 8. 48-hour $PM_{2.5}$ forecasting comparison between SARIMAX Recursive and SARIMAX Dynamic models.

Figure 9 summarizes the comparative MAPE performance of the evaluated forecasting approaches on the shared holdout window.

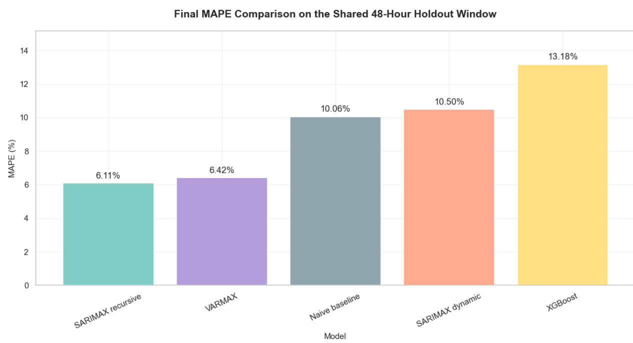

Fig. 9. MAPE comparison across forecasting models on the 48-hour holdout window.

The forecasting experiments showed that SARIMAX with the recursive mode has achieved the overall best forecasting performance across evaluated models. As shown in Table V, it has produced the lowest MAPE score, as well as the lowest RMSE, MAPE, and sMAPE, and achieved the highest R^2 on the Lahore holdout forecasting with a shared 48-hour window.

TABLE V
FORECASTING PERFORMANCE COMPARISON

Model	MAE	RMSE	MAPE (%)	sMAPE (%)	R^2
SARIMAX recursive	9.991	16.096	6.108%	5.871	0.903
VARMAX	11.263	18.515	6.419%	6.303	0.872
Naive baseline	16.273	20.848	10.063%	9.893	0.838
SARIMAX dynamic	17.876	23.448	10.499%	10.594	0.795
XGBoost	20.076	24.082	13.184%	12.170	0.783

As SARIMAX was the best out of all forecasting models performed on this time series, it was selected for the final 7-day hourly $PM_{2.5}$ forecasts for all the cities included in the dataset. Figure 10 shows the final 7-day $PM_{2.5}$ forecast generated for Lahore using the recursive SARIMAX model.

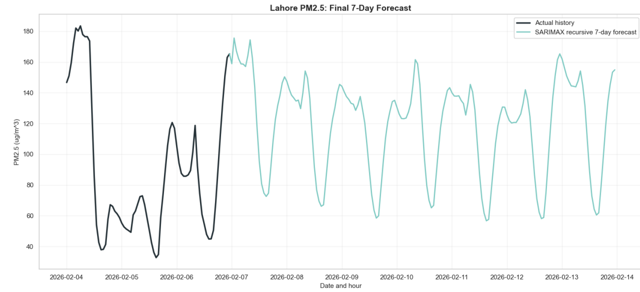

Fig. 10. Final 7-day $PM_{2.5}$ forecast for Lahore using the SARIMAX Recursive model.

To further validate forecasting reliability, actual $PM_{2.5}$ observations were obtained from the official WAQI platform and compared against the generated forecasts for the cities used in these analyses.

VI. Results

Figure 11 summarizes the final MAPE score obtained by the SARIMAX model after forecasting for all the cities and comparing with the actual observations later obtained from the *World Air Quality Index (WAQI)* platform. As observed, the model performed satisfactorily on cities such as Dhaka, London, Lahore, Delhi, Jakarta, Sao Paulo, and New York, achieving a MAPE score of less than 26%. However, for Beijing, it attained a MAPE score of almost 70%, possibly indicating that some other external factors were present that did not appear in the dataset, and those factors were possibly specific to Beijing or the territory of China, causing the city's air quality pattern to deviate from the SARIMAX forecasts.

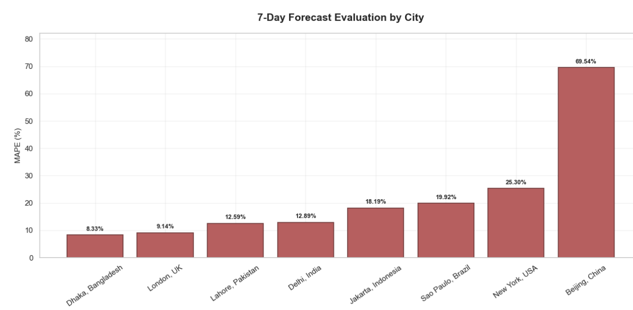

Fig. 11. 7-day $PM_{2.5}$ forecast evaluation by city using MAPE.

Figures 12, 14, and 13 illustrate the SARIMAX forecasts compared with the real observations obtained for the forecasted period for the cities Lahore, London, and Beijing.

Visually, it can be observed that the forecasted and actual series are fairly close; however, the MAPE score for Beijing is almost 70%, which appeared this high because MAPE divides the error by the real $PM_{2.5}$ value. When the real value becomes very small, even a small prediction difference can turn into a very large percentage error.

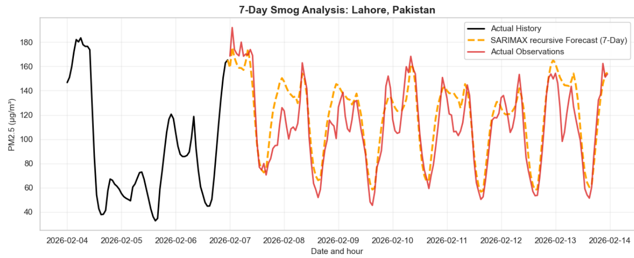

Fig. 12. 7-day $PM_{2.5}$ forecast versus WAQI actual observations for Lahore.

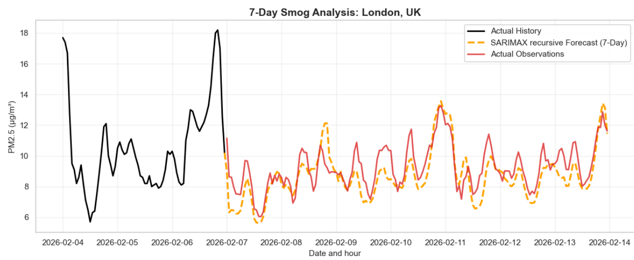

Fig. 13. 7-day $PM_{2.5}$ forecast versus WAQI actual observations for London.

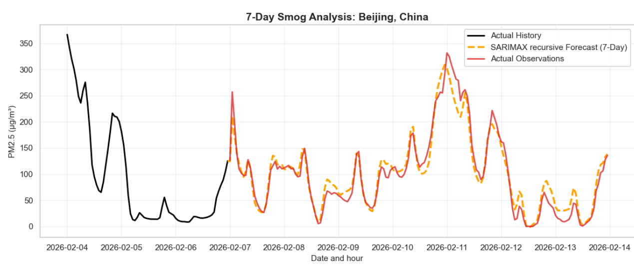

Fig. 14. 7-day $PM_{2.5}$ forecast versus WAQI actual observations for Beijing.

For such a reason, other metrics that were considered during these analyses may help derive correct insights from the results. Table VI shows all metric scores obtained for all the cities' forecasts compared with the actual observations retrieved from WAQI for the 7-day forecasted horizon.

TABLE VI
7-DAY FORECAST EVALUATION METRICS BY CITY USING RECURSIVE SARIMAX

City	MAE	RMSE	MAPE (%)	sMAPE (%)	R^2
Dhaka, Bangladesh	6.217	8.055	8.333	8.084	0.815
London, UK	0.846	1.032	9.138	9.537	0.468
Lahore, Pakistan	12.915	16.163	12.585	11.614	0.771
Delhi, India	11.712	14.682	12.889	11.951	0.781
Jakarta, Indonesia	6.982	8.938	18.186	16.388	0.866
Sao Paulo, Brazil	2.962	3.763	19.920	17.236	0.509
New York, USA	4.862	5.471	25.302	30.525	0.449
Beijing, China	13.466	17.615	69.540	25.939	0.947

Here, it can clearly be observed that MAE and RMSE scores for Beijing remain low, like for Lahore, which had a much smaller final MAPE value, showing that MAPE can become inflated when actual $PM_{2.5}$ values are very small. This brings another important insight from this work: that *no single metric can be fully trusted throughout the whole process of analysis, and other benchmarks should also be considered.*

Overall, the results show that SARIMAX Recursive can capture short-term daily smog cycles and general $PM_{2.5}$ movement patterns. However, its performance varies across cities because pollution behavior depends on local dynamics, sudden atmospheric changes, and the availability of representative historical patterns.

VII. DISCUSSION

The forecasting models were generally successful at capturing the strong seasonal structure and recurring patterns observed during the exploratory analysis. The ACF and PACF, as well as the previously applied seasonal decomposition, indicated a strong 24-hour seasonal structure, which explains why statistical models performed well.

Recursive SARIMAX performed better than its dynamic counterpart, as the previously generated values preserved the local short-term temporal structure more effectively. VARMAX performed similarly to the recursive SARIMAX, whereas XGBoost was the worst performer, possibly because it requires additional feature engineering, a larger dataset, or more advanced tuning to capture long sequential dependencies in strongly seasonal time series data.

The best-performing model was Recursive SARIMAX; therefore, it was chosen for predicting the series. Cities such as Dhaka, London, Lahore, and Delhi achieved low MAPE scores because $PM_{2.5}$ patterns were more stable in the actual series observed for the forecast period, as retrieved from WAQI. But Table VI showed that other metrics should also be considered when analyzing the overall performance of SARIMAX forecasts, because extreme values in pollutant observations can cause inflated error scores, as in the case of MAPE for Beijing.

Overall, the experiments conducted to forecast $PM_{2.5}$ concentrations from the 14th to the 21st of February 2026 yielded decent results, indicating that classical time-series forecasting approaches are highly competitive in environmental prediction when a strong seasonal structure is present.

VIII. CONCLUSION

This project focused on forecasting $PM_{2.5}$ concentrations during the smog season, started in October 2025 to February 2026, for cities such as Beijing, Delhi, Dhaka, Jakarta, Lahore, London, New York, and Sao Paulo. During the exploratory analysis, the *Global Air Pollution Dataset (2025-2026)* [5] was analyzed to better understand multivariate pollutant interaction and behavior.

Several forecasting approaches, such as `SARIMAX` with recursive and dynamic modes, `VARMAX`, and `XGBoost`, alongside a `Naive` (last value) Baseline, were implemented and evaluated on the 48-hour forecasting window for Lahore. The results were evaluated using different error metrics such as `MAE`, `RMSE`, `MAPE`, and `sMAPE`.

Among all the models, the Recursive SARIMAX was selected as the best one to conduct the final 7-day forecast for all cities present in the dataset. Afterward, the actual observations for the third week of February were retrieved from the official *WAQI* platform to compare the performance of Recursive SARIMAX. Overall, the comparison showed a good performance behavior attained by the model, and it could fairly depict the main pollutant patterns in different cities.

The objective of this project was successfully met, with the analysis, comparisons, and forecasting for the short-term $PM_{2.5}$ pollutant concentration levels using time series and machine learning models.

The implementation details, source code, and forecasting experiments developed for this project are publicly available in the project repository. [7]

IX. Future Work

For future work, additional variables such as temperature, humidity, wind speed, and atmospheric pressure may be considered as external factors crucial for forecasting and improving models to capture sudden pollution changes and extreme smog events.

For longer-term historical data, several deep learning architectures can be considered, such as LSTMs, GRUs, or transformer-based models, to capture long-term sequential dependencies and nonlinear behavior.

Finally, extend the project to a real-time deployment and integrate live air-quality APIs to support continuous forecasting. These enhancements can serve as better drivers for decision-making for environmental monitoring and public health applications.

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